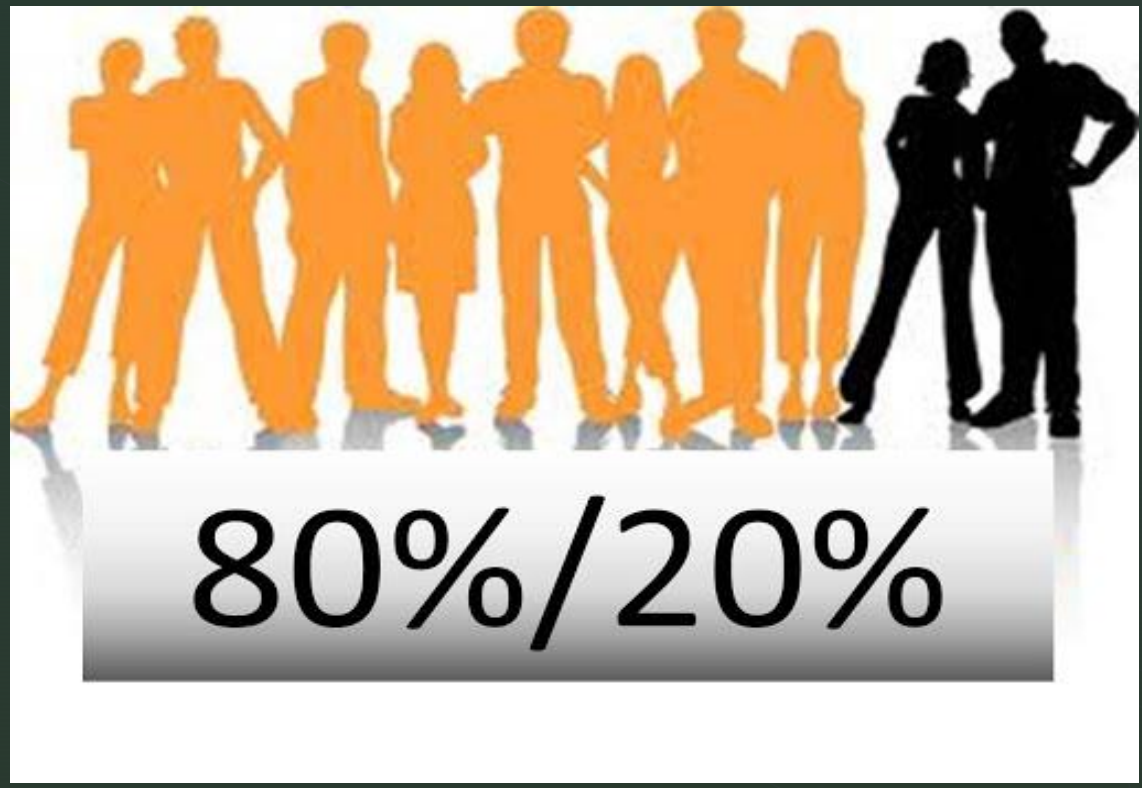
Measures of Location

Main Content

- Quartile
- Decile
- Percentile

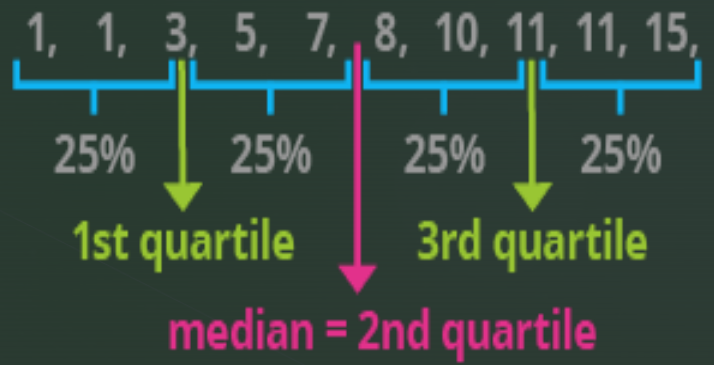


What is Quartile ?

Values that divide a list of numbers into quarters.

- 1st Quartile = 25% of distribution.
- 2nd Quartile = 50% of distribution.
- 3rd Quartile = 75% of distribution.

Dividing the data set into 4 equal parts.



What is Decile ?

Dividing the data set into 10 equal parts.

- 1st Decile = 10% of distribution.
- 2nd Decile = 20% of distribution.

……..

……

- 9th Decile = 90% of distribution.



What is Percentile ?

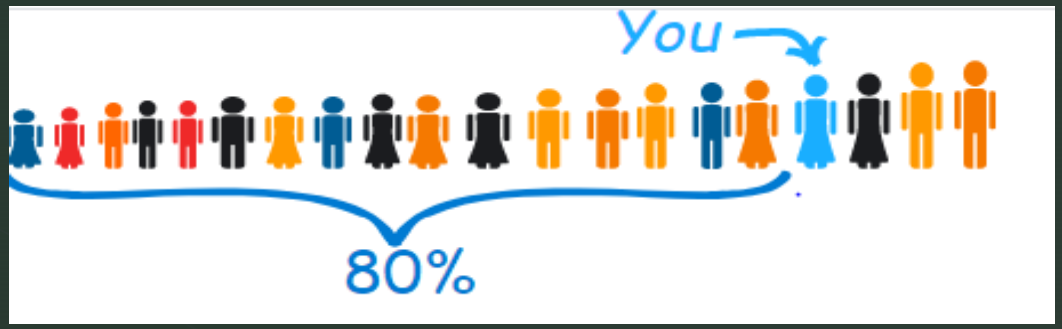
Dividing the data set into 100 equal parts.

- 1st Percentile = 1% of distribution.
- 2nd Percentile = 2% of distribution.

……..

……

- 98th Percentile = 98% of distribution.
- 99th Percentile = 99% of distribution.



Quartile, Percentile and Decile for ungrouped data

8,10,15,13,3,6,4,12,9,7,5,1,11,2,14

1. Arrange the data from lowest to highest (ascending) order.

1,2,3,4,5,6,7,8,9,10,11,12,13,14,15

N = 15 (total number of data set)

- Q1 = 4th value=4
- Q2 = 8th value =8
- Q3 = 12th value=12

For Quartile, $Q_i=\frac{i(n+1)}{4}$

For Decile, $D_i=\frac{i(n+1)}{10}$

For Percentile, $P_i=\frac{i(n+1)}{100}$

Here i=1,2,3….

n= Total no of observation

Calculate the 3rd quartile, 6th decile and 70th percentile

28, 17, 12, 25,26, 19, 13, 27, 21,16

We first arrange in ascending order

12, 13, 16, 17, 19, 21, 25,26, 27, 28

Q_3=$\frac{3(10+1)}{4}$ th. Value = 8.25th value

= 8th $value$ + 0.25 (9th $value$ − 8th $value$)= 26 + 0.25(27 − 26) = 26.25

So, 75% observation is 26.25.

Calculate the 3rd quartile, 6th decile and 70th percentile

28, 17, 12, 25,26, 19, 13, 27, 21,16

We first arrange in ascending order

12, 13, 16, 17, 19, 21, 25,26, 27, 28

$D_6 = \frac{6(10+1)}{10} th\ value$ = 6.6$th\ value$

$D6$ = 6$th\ value$ + 0.6 (7$th\ value$ − 6$th\ value$) = 21 + 0.6 (25 − 21) = 23.4

60% observation is 23.4

70th Percentile try yourself

Quartile, Percentile and Decile for grouped data

For a grouped frequency distribution, we use n instead of n+1

$$Q_1 = l + (\frac{\frac{n}{4} - cf}{f}) \times c$$

$$Q_2 = l + \frac{(\frac{n}{2} - cf}{f}) \times c$$

$$Q_3 = l + (\frac{\frac{3n}{4} - cf)}{f} \times c$$

Where l = lower boundary of the class in which $Q_1 lies$

cf = cumulative frequency of the class preceeding this class

f = frequency of the class in which Q_1 *lies*

$c = width\ of\ the\ class\ interval$

Continue….

$$D_r = l + (\frac{\frac{rn}{10} - cf)}{f} \times c$$

$$P_r = l + (\frac{\frac{rn}{100} - cf)}{f} \times c$$

Example

C I	f	c f
600-700	40	40
700-800	68	108
800-900	86	194
900-1000	120	314
1000-1100	90	404
1100-1200	40	444
1200-1300	26	470

$$\frac{470}{4} = 117.5$$

We take 800 -900 as the class in which $Q_1 lies$

$$Q_1 = 800 + \left(\frac{117.5 - 108}{86}\right) \times 100$$

$$= 811.04$$

Continue….

$$Q_2 = 900 + \frac{(235 - 194}{120}) \times 100 = 934.17$$

$$\frac{470}{2} = 235$$

$$\frac{3(470)}{4} = 352.5$$

$$Q_3 = 1000 + (\frac{352.5-314}{90}) \times 100 = 1042.78$$

$$\frac{3(470)}{10} = 141$$

$$D_3 = 800 + \left(\frac{141 - 108}{86}\right) \times 100 = 838.37$$

$$\frac{95(470)}{100} = 446.5$$

$$P_{95} = 1200 + \left(\frac{446.5 - 444}{26}\right) \times 100 = 1209.62$$

Box-and-Whisker Plots

- Five Number Summary:
  - Lowest Value
  - Q1
  - Median
  - Q3
  - Highest Value

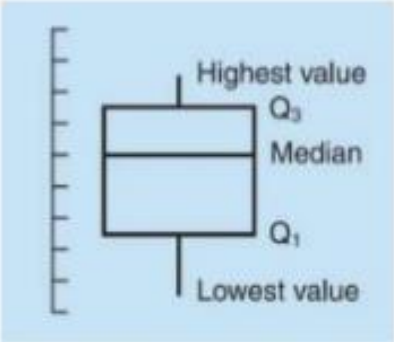
The five number summary is used to create a graphic sketch of the data called a box-and- whisker plot.

Why box-plot?

- A box-and-whisker plot is a visual representation a five-number summary.
- Provides a graphic display of the spread of the data about the median and a quick overall summary of symmetry/skewness
- The box shows the middle half of the data. One quarter of the data is located along each whisker.
- If the median line is near one end of the box, the data are skewed toward the other end of the box.
- If the median is approximately centered in the box and the whiskers are about the same length, the data is approximately symmetric.
- For identifying outlier.

How to draw box plot ?

1. Find the five number summary
2. Draw a vertical scale to include the lowest and highest data values
3. To the right of the scale, draw a box from Q1 to Q3
4. Include a solid line through the box at the median level
5. Draw vertical lines, called *whiskers*, from Q1 to the lowest value and from Q3 to the highest value.



Box-and-Whisker Plot

Figure 3-6

Inner fence and Outer fence

Inner fence:

Q_1 -1.5 (IQR), Q_3 +1.5 (IQR)

Outer fence :

Q_3 -3 (IQR), Q_3 +3 (IQR)

Where IQR =Inter Quartile Range=Q_3-Q_1

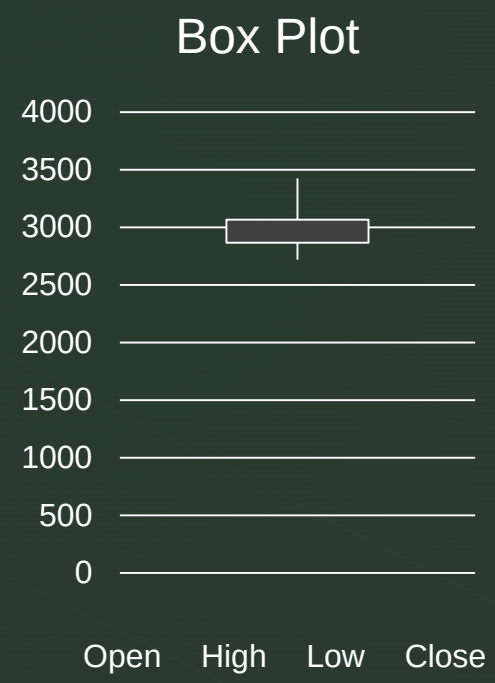
Example

2890 2930 3425 2860 2960 3060 2880 2765 2720 2900 2950 3260

After rearranged in ascending order

2720 2765 2860 2880 2890 2900 2930 2950 2960 3060 3260 3425

- Five Number Summary:
  - Lowest Value = 2720
  - Q1 =2865
  - Median = 2915
  - Q3 = 3035
  - Highest Value = 3425



Detecting outlier

Inner fence:

Q_1 -1.5 (IQR)= 2865-1.5*170=2610

Q_3 +1.5 (IQR)= 3035+1.5*170= 3290

Outer fence :

Q_3 -3 (IQR)=3035-3*170=2525

Q_3 +3 (IQR)=3035+3*170=3545

Where IQR =Inter Quartile Range=Q_3-Q_1=3035-2865=170

The values which are not in the range of inner fence(2610-3290) they are called outlier.

So 3425 is the only outlier.


Thank You !